

FILIPPO MUSCHERÀ

Computer Engineer

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📍 Rome, Italy

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🐙 FilippoMuschera



PROFILE

M.Sc. in Computer Engineering with study abroad experience in Sweden. My academic path and thesis work focus on Cloud and Edge Computing systems, distributed architectures, and Machine Learning. I address problems across different abstraction levels, ranging from the design of distributed architectures to operating system-level optimization.

EDUCATION

M.Sc. in Computer Engineering

Università degli Studi di Roma "Tor Vergata"

📅 Mar 2023 – Mar 25, 2026

📍 Rome, Italy

Final grade: 110/110 cum laude

Master's Thesis: Request Distribution in Multi-Architecture Function-as-a-Service Systems (Advisors: Prof. V. Cardellini, Prof. G. R. Russo)

- Extended the **FaaS** platform Serverledge to natively support and manage heterogeneous compute nodes (x86-64 and ARM64) in Edge-Cloud environments.
- Designed an Architecture-Aware Load Balancer using **Reinforcement Learning** algorithms (Multi-Armed Bandit, UCB1, LinUCB) for dynamic function scheduling based on architectural affinity, response times, and memory utilization.
- Integrated a **Consistent Hashing** mechanism to optimize request distribution, maximizing resource affinity and reducing cold start latencies. Automated the final deployment on Google Cloud using **Terraform** and **Ansible**.

B.Sc. in Computer Engineering

Università degli Studi di Roma "Tor Vergata"

📅 Sept 2018 – Feb 2023

📍 Rome, Italy

- Final grade: 108/110

EXPERIENCE

Erasmus+ Programme

Blekinge Tekniska Högskola (BTH)

📅 Jan 2025 – Jun 2025

📍 Karlskrona, Sweden

- Completed a **6-month** study period within the M.Sc. in Computer Engineering program.
- Focused on courses in **Compiler Design, Research Methodologies, Data-Driven Development and Decision Support Systems**.
- Worked on several group projects, enhancing my communication skills, English proficiency, and teamwork abilities in an **international environment**.

OPEN SOURCE CONTRIBUTIONS

Linux Kernel (torvalds/linux)

- Contributed patches to the hardware monitoring and I2C subsystems, focusing on code modernization and memory safety.
- Migrated legacy drivers to the *managed* device API (`devm_*`), streamlining I/O resource management, reducing boilerplate code, and simplifying error handling paths.
- Refactored `sysfs` callback functions using `sysfs_emit` to enforce strict boundary checks and **protect against buffer overflows**.

Serverledge (FaaS Platform)

- Optimized the container pool management (idle/busy pools) by replacing linked lists with Go slices, leveraging *memory locality* and in-place filtering.
- Profiled the optimization via Go benchmarks, demonstrating a reduction in memory allocations and a **50-80% improvement in execution time** for iterative pool operations.

PROJECTS

AI & Machine Learning

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- Implemented and compared various **Machine Learning** and **Convolutional Neural Network (CNN)** models to classify a compressed version of the images in the MNIST-Pneumonia database, reaching 98.3% accuracy. Available [on GitHub](#)

Data Driven Development

📍 Blekinge Tekniska Högskola

- Applied **Transfer Learning** and a **Transformer** model for **Sentiment Analysis** on Airbnb reviews, then developed regression models for listing price prediction. Conducted **data analysis and visualization** to interpret model outputs and findings, available [on GitHub](#)

Cloud and Distributed Systems

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- Designed and implemented a **distributed Key-Value Store** service in **Go**, supporting Sequential and Causal Consistency protocols. Deployed and tested the solution using **Docker Compose** on **AWS EC2** instances. Available [on GitHub](#)

Malware Analysis

📍 Università degli Studi di Roma "Tor Vergata"

- Executed **Malware Reverse Engineering** (static and dynamic analysis) on a Windows executable. The analysis identified a two-stage attack involving code injection via Asynchronous Procedure Call and subsequent runtime de-obfuscation of the shellcode to reveal its Command&Control (C2) protocol. Available [here](#)

Software Testing

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- Designed and executed **Unit** and **Integration Tests** for two Apache software projects, leveraging tools like PIT Mutation Testing and JaCoCo for coverage analysis. Automated the testing workflow using **GitHub Actions**. Available [here](#)

SKILLS

Programming Languages & Frameworks

C Python Go Java Kotlin JUnit Jetpack Compose

Data Science & Machine Learning

TensorFlow NumPy Pandas Matplotlib CUDA/cuDNN

Cloud & Tools

Docker Docker Compose Terraform Ansible AWS EC2 Google Cloud Compute Engine Git
GitHub Actions Linux Network Simulation / GNS3 MySQL Ghidra Bash Scripting

LANGUAGES

- Italian – Native
- English – Full Professional Proficiency